4.3.13 Grades of Concrete

Concrete grades are denoted by M_{10}, M_{20}, M_{30} according to their compressive strength. The 'M' denotes mix design of concrete followed by the compressive strength number in N/mm^2.

'Mix' is the respective ingredient proportions which are cement, fine aggregate and coarse aggregate.

If we mention M_{10} concrete, it means that, the concrete has 10 N/mm^2 characteristic compressive strength after 28 days.



ACTIVITY 6

Collect pictures of coloured concrete buildings and various concrete construction and prepare an album.

Note: The minimum grade of concrete for plain cement concrete (PCC) is M_{15} and for reinforced cement concrete (RCC) is $M_{20.}$

Grade	Proportion of concrete	Characteristic Compressive Strength (N/mm^2)
M_5	1:5:10	5 N/mm^2
$M_{7.5}$	1:4:8	7.5 N/mm^2
M_{10}	1:3:6	10 N/mm^2
M_{15}	1:2:4	15 N/mm^2
M_{20}	1:1 ½ : 3	20 N/mm^2
M_{25}	1: 1: 2	25 N/mm^2
M_{30}	1: 1: 3	30 N/mm^2

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